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Physical Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

Chemical Kinetics

English Medium

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CHEMICAL KINETICS

1. INTRODUCTION

Chemical kinetics deals with the rates of chemical processes. Any chemical process may be broken down into a sequence of one or more single-step processes known either as elementary processes, elementary reactions, or elementary steps. Elementary reactions usually involve either a single reactive collision between two molecules, which we refer to as a bimolecular step, or dissociation/ isomerisation of a single reactant molecule, which we refer to as a unimolecular step. Very rarely, under conditions of extremely high pressure, a termolecular step may occur, which involves simultaneous collision of three reactant molecules. An important point to recognise is that, many reactions that are written as a single reaction equation, in actual fact, consist of a series of elementary steps. This will become extremely important as we learn more about the theory of chemical reaction rates.

As a general rule, elementary processes involve a transition between two atomic or molecular states separated by a potential barrier. The potential barrier constitutes the activation energy of the process, and determines the rate at which it occurs. When the barrier is low, the thermal energy of the reactants will generally be high enough to surmount the barrier and move over to products, and the reaction will be fast. However, when the barrier is high, only a few reactants will have sufficient energy, and the reaction will be much slower. The presence of a potential barrier to reaction is also the source of the temperature dependence of reaction rates.

The huge variety of chemical species, types of reaction, and the accompanying potential energy surfaces involved means that the time scale over which chemical reactions occur covers many orders of magnitude, from very slow reactions, such as iron rusting, to extremely fast reactions, such as the electron transfer processes involved in many biological systems or the combustion reactions occurring in flames.

A study into the kinetics of a chemical reaction is usually carried out with one or both of two main goals in mind :

(i) Analysis of the sequence of elementary steps giving rise to the overall reaction. i.e. the reaction mechanism.

(ii) Determination of the absolute rate of the reaction and/or its individual elementary steps.

2. CLASSIFICATION OF REACTION

(i) There are certain reactions which are too slow. **Ex.** Rusting of Iron, weathering of rocks.

(ii) Instantaneous reactions i.e. too fast. **Ex.** Detonation of explosives, acid-base neutralization, precipitation of AgCl by NaCl and $AgNO_3$.

(iii) Neither too fast nor too slow. **Ex.** Combination of H_2 and Cl_2 in presence of light, hydrolysis of ethyl acetate catalysed by acid, decomposition of Azomethane.

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3. Types of Rates of chemical reaction :

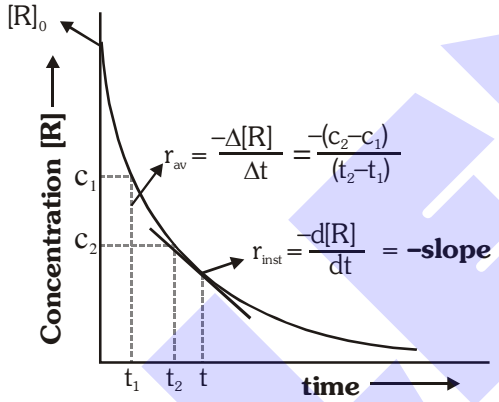
For a reaction R $\longrightarrow$ P ,

- **Average rate** $= \frac{\text{Total change in concentration}}{\text{Total time taken}} = \frac{|\Delta c|}{\Delta t} = -\frac{\Delta[\text{Reactant}]}{\Delta t} = \frac{\Delta[\text{Product}]}{\Delta t}$
- **Instantaneous rate** : Rate of reaction at a particular instant.

$$\mathbf{R_{instantaneous}} = \lim_{\Delta t \to 0}\left[\frac{|\Delta c|}{\Delta t}\right] = \left|\frac{dc}{dt}\right| = -\frac{d[R]}{dt} = \frac{d[P]}{dt}$$

Instantaneous rate can be determined from slope of a tangent at time t on curve drawn for concentration versus time.

3.1 Initial Rate : Instantaneous rate at 't = 0' is called initial rate [Slope of tangent at t = 0].



3.2 Reaction rates and stoichiometry :

We have seen that for stoichiometrically simple reactions of the type A → B, the rate can be either expressed in terms of the decrease in reactant concentration with time, $-\Delta[A]/\Delta t$ or the increase in product concentration with time, $\Delta[B]/\Delta t$. For more complex reactions, we must be careful in writing the rate expressions. Consider for example, the reaction,

2 A $\longrightarrow$ B

Two moles of A disappear for each mole of B that forms – that is, the rate of disappearance of A is twice as fast as the rate of appearance of B. We write the overall rate of reaction as either

$$\text{Rate} = -\frac{1}{2}\frac{\Delta[A]}{\Delta t} = \frac{\Delta[B]}{\Delta t}$$

In general, for the reaction, **aA + bB $\longrightarrow$ cC + dD**

the rate is given by

$$\text{Rate} = -\frac{1}{a}\frac{\Delta[A]}{\Delta t} = -\frac{1}{b}\frac{\Delta[B]}{\Delta t} = \frac{1}{c}\frac{\Delta[C]}{\Delta t} = \frac{1}{d}\frac{\Delta[D]}{\Delta t}$$

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Ex. **For the reaction in terms : $N_2 + 3H_2 \longrightarrow 2NH_3$**

Rate of reaction in terms of $N_2 = -\frac{d[N_2]}{dt}$ = rate of disappearance of N_2

Rate of reaction in terms of $H_2 = -\frac{d[H_2]}{dt}$ = rate of disappearance of H_2

Rate of reaction in terms of $NH_3 = \frac{d[NH_3]}{dt}$ = rate of appearance of NH_3

These rates are not all equal. Therefore, by convention, the rate of a reaction is defined as

- **Rate of reaction** $= -\frac{d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$

Note: The value of rate of reaction is dependent on the stoichiometric coefficients used in the reaction while the rate of increase or decrease in amount of any species will be fixed value under given conditions.

Ex.1 **The following reaction was studied in a closed vessel**

$$2N_2O_5(g) \longrightarrow 4NO_2(g) + O_2(g)$$

It was found that the concentration of NO_2 increases by 2.0×10^{-2} mol L^{-1} in five seconds. Calculate.

(i) **The rate of reaction**

(ii) **The rate of decrease of concentration of N_2O_5**

Sol. **(i)** Rate of reaction $= \frac{1}{4}\frac{d[NO_2]}{dt}$

But $\frac{d[NO_2]}{dt} = \frac{2.0 \times 10^{-2} \text{ mol } L^{-1}}{5s} = 4 \times 10^{-3} \text{ mol } L^{-1} s^{-1}$

Rate of reaction $\frac{1}{4} \times 4 \times 10^{-3} = 10^{-3} \text{ mol } L^{-1} s^{-1}$

(ii) Rate of decrease of conc. of N_2O_5

$= -\frac{d[N_2O_5]}{dt} = -\frac{1}{2} \times$ Rate of formation of $NO_2 = +\frac{1}{2}\frac{d[NO_2]}{dt}$

$= +\frac{1}{2} \times 4 \times 10^{-3} = 2 \times 10^{-3} \text{ mol } L^{-1} s^{-1}$

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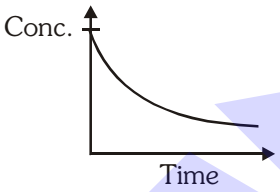
3.3 FACTORS AFFECTING RATE OF CHEMICAL REACTION

(i) Concentration **(ii)** Temperature
(iii) Nature of reactants and products **(iv)** Catalyst
(v) pH of the solution **(vi)** Dielectric constant of the medium.
(vii) Radiations / light **(viii)** Pressure
(ix) Electrical and magnetic field.

The first four factors generally affect rate of almost all reactions while other factors are specific to some reactions only.

(i) Effect of concentration :

For most of the reactions, rate depends on concentration of reactants. So rate of reaction decreases with passage of time, since concentration of reactants decreases .



(ii) Effect of temperature : Generally rate of reaction increases on increasing temperature.

(iii) Effect of nature of reactants and products :

(a) Physical state of reactants : Gaseous state > Liquid state > Solid state
(Decreasing order of rate of reaction)

Because collisions in gaseous systems are more effective than condensed systems (solid & liquid).

(b) Physical size of reactants : In heterogeneous reactions, as we decreases the particle size, rate of reaction increases since surface area increases.

(c) Chemical nature of reactants :

- If more bonds are to be broken, the rate of reaction will be slow.
- Similarly if bond strength in reactants is more, rate of reaction will be slow.

(iv) Effect of Catalyst :

- Presence of catalyst or positive catalyst lowers down the activation energy hence increases the rate of reaction.
- Presence of inhibitor or negative catalyst increases activation energy hence decreases the rate of reaction.

(v) Effect of pH of solution : Ex. $[Fe(CN)_6]^{4-} \xrightarrow{(Tl^{3+})} [Fe(CN)_6]^{3-}$

This reaction takes place with appreciable rate in acidic medium, but does not take place in basic medium.

(vi) Effect of dielectric constant of the medium : More is the dielectric constant of the medium greater will be the rate of ionic reactions.

(vii) Effect of radiations / light : Radiations are useful for photochemical reactions.

(viii) Effect of pressure : Pressure is important factor for gaseous reactions.

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(ix) **Effect of electrical & magnetic field :** Electric and magnetic fields are rate determining factors if a reaction involves polar species.

4. RATE LAW (DEPENDENCE OF RATE ON CONCENTRATION OF REACTANTS) :

The representation of rate of reaction in terms of the concentration of the reactants is called the rate law. It can only be established by experiments.

Generally rate law expressions are not simple and these may differ for the same reaction on conditions under which the reaction is being carried out. But for large number of reactions starting with pure reactants we can obtain simple rate laws. For these reactions :

Rate $\propto$ (conc.)^{order}

Rate = K (conc.)^{order} **This is the differential rate equation or rate expression.**

Where K = Rate constant = specific reaction rate = rate of reaction when concentration is unity

unit of K = (conc)$^{1-\text{ order}}$ time^{-1}

Note: Value of K is a constant for a given reaction, depending only on temperature and catalyst use.

4.1 Order of reaction :

Let there be a reaction , $m_1A + m_2B \longrightarrow$ products.

Now, if on the basis of experiment, we find that

$$R \propto [A]^p [B]^q$$

where p may or may not be equal to m_1 and similarily q may or may not be equal to m_2.

p is order of reaction with respect to reactant A and q is order of reaction with respect to reactant B and (p + q) is **overall order of the reaction.**

Note: Order of a reaction can be 'zero', any whole number, fractional number or even be negative with respect to a particular reactant.

❖ **Examples showing different values of order of reactions :**

	Reaction	Rate law	Order
(i)	$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$	$R = K[N_2O_5]^1$	1
(ii)	$5Br^-(aq) + BrO_3^-(aq) + 6H^+(aq) \rightarrow 3Br_2(\ell) + 3H_2O(\ell)$	$R = K[Br^-][BrO_3^-][H^+]^2$	1 + 1 + 2 = 4
(iii)	H_2 (Para) $\rightarrow H_2$ (ortho)	$R = K[H_{2\ (Para)}]^{3/2}$	3/2
(iv)	$NO_2(g) + CO(g) \rightarrow NO(g) + CO_2(g)$	$R = K[NO_2]^2[CO]^o$	2 + 0 = 2
(v)	$2O_3(g) \rightarrow 3O_2(g)$	$R = K[O_3]^2[O_2]^{-1}$	2 – 1 = 1
(vi)	$H_2 + Cl_2 \xrightarrow{h\nu} 2HCl$	$R = K[H_2]^o[Cl_2]^o$	0 + 0 = 0

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The reaction (ii) does not take place in one single step. It is almost impossible for all the 12 molecules of the reactants to be in a state of encounter simultaneously. Such a reaction is called ***Complex reaction*** and takes places in a sequence of a number of ***Elementary reactions***. For an elementary reaction, the sum of stoichiometric coefficients of reactants = order of the reaction. But for complex reactions, order is to be experimentally calculated.

4.2 Molecularity of reaction :

The number of molecules that react in an elementary step is the molecularity of the elementary reaction. Molecularity is defined only for the elementary reactions and not for complex reactions. No elementary reactions involving more than three molecules are known, because of very low probability of simultaneous collision of more than three molecules.

The rate law for the elementary reaction

$aA + bB \longrightarrow$ Products, Rate = $k[A]^a[B]^b$, where a + b = 1, 2 or 3

For an elementary reaction, the orders in the rate law equal the coefficients of the reactants. It must be noted that the order is defined for complex as well as elementary reactions and is always experimentally calculated from the mechanism of the reaction, usually by the slowest step of the mechanism known as rate determining step (RDS) of the reaction.

Comparision between molecularity and order of reaction

	Molecularity of Reaction		***Order of Reaction***
1	It is defined as the no. of molecules of reactant taking part in a chemical reaction. $NH_4NO_2 \rightarrow N_2 + 2H_2O$, molecularity = 1	1	It is defined as the sum of the power of concentration terms that appear in rate law. $NH_4NO_2 \rightarrow N_2 + 2H_2O$. Rate = $k[NH_4NO_2]$, order =1
2	It is always a whole number. It can neither be zero nor fractional.	2	It may be zero, fractional or integer.
3	It is derived from RDS in the mechanism of reaction	3	It is derived from rate expression.
4	It is theoretical value.	4	It is experimental value.
5	Reactions with molecularity > 3 are rare.	5	Reactions with order of reaction > 4 are also rare.
6	Molecularity is independent of pressure and temperature.	6	Order of reaction may depend upon pressure and temperature.

5. INTEGRATED RATE LAW :

For a single reactant reaction where the chemical equation has the form

$A \rightarrow$ products

and the rate law is assumed to be of the form

Rate = $-d[A]/dt = k[A]^m$

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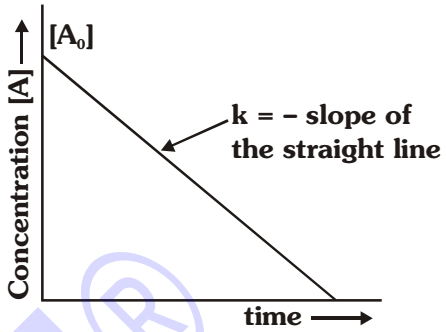
Where m is the order of the reaction with respect to substance A. Three important cases can be treated : m = 0, m = 1, and m = 2. These are called ***zeroth order***, ***first order***, and ***second order***, respectively.

5.1 Zero - order reaction :

For a reaction where the chemical equation has the form

A → products

and the rate law is assumed to be of the form



$$\textbf{Rate} = -\frac{d[A]}{dt} = k[A]^0$$

$$-\int_{c_0}^{c_t} d[A] = k\int_0^t dt$$

$$k = \frac{C_0 - C_t}{t}$$

or $\mathbf{kt = C_0 - C_t}$ or $C_t = C_0 - kt$

- Unit of K is same as that of Rate = mol lit^{-1} sec^{-1}.
- Time for completion $= \frac{C_0}{k}$
- $t_{1/2}$ (half life period)

At $t_{1/2}$, $C_t = \frac{C_0}{2}$,

So $kt_{1/2} = \frac{C_0}{2}$ $\Rightarrow$ $t_{1/2} = \frac{C_0}{2k}$

$\therefore$ $t_{1/2} \propto C_0$

❖ Examples of Zero order reactions :

Generally decomposition of gases on metal surfaces at high concentrations follow zero order kinetics as rate depends on surface area of catalyst.

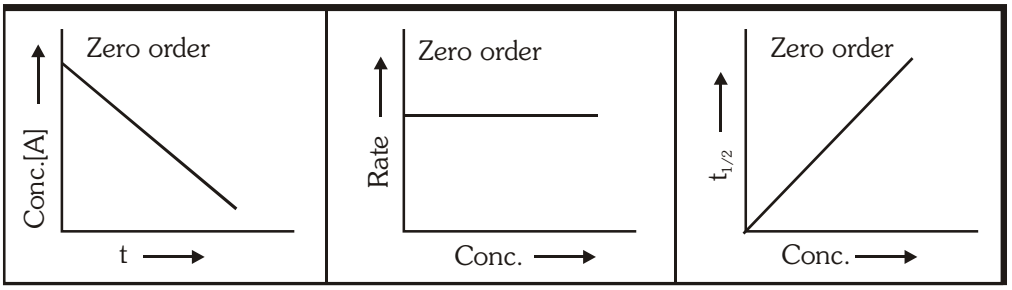
$2PH_3(g) \xrightarrow{Ni} 2P(s) + 3H_2(g)$ **Rate = K $[PH_3]^0$**

$2HI(g) \xrightarrow{Au} H_2(g) + I_2(g)$ **Rate = K $[HI]^0$**

$2NH_3(g) \xrightarrow{Pt} N_2(g) + 3H_2(g)$ **Rate = K $[NH_3]^0$**

$H_2(g) + Cl_2(g) \xrightarrow{hv} 2HCl(g)$ **Rate = K $[H_2]^0$ $[Cl_2]^0$**

❖ Graphs :



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Ex.2 For the zero order reaction : A → P , K = 10^{-2} (mol/litre) sec^{-1} If initial concentration of A is 0.3M , then find concentration of A left at 10 sec.

(A) 0 M **(B) 0.2 M** **(C) 0.1 M** **(D) 0.15 M**

Sol. (B)

$[A]_t = [A]_0 - Kt = 0.3 - 10^{-2} \times 10 = 0.2$ M

5.2 First order reaction :

Consider a first order reaction with single reactant.

	A ⟶	**Products**
t = 0	a	0
t = t	a–x	

Rate = – d[A]/dt = k $[A]^1$

$\therefore \quad \frac{dx}{dt} = k(a-x)^1 \quad$ or $\quad \frac{dx}{a-x} = kdt.$

- **On solving** $k = \frac{2.303}{t} \log \frac{a}{a-x} = \frac{2.303}{t} \log \frac{C_0}{C_t}$

$K = \boxed{\frac{2.303}{t} \log \frac{C_0}{C_t}}$ $\boxed{\text{Wilhemy formula : } C_t = C_0 e^{-kt}}$ $\boxed{\text{Interval formula : } k = \frac{2.303}{(t_2 - t_1)} \log \frac{C_1}{C_2}}$

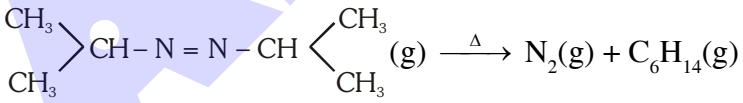
- Unit of k = sec^{-1}, min^{-1}, etc.
- **Half-life time ($t_{1/2}$) :**

$$k = \frac{2.303}{t_{1/2}} \log \frac{2C_0}{C_0} \Rightarrow t_{1/2} = \frac{2.303 \log 2}{k} = \frac{\ln 2}{k} = \frac{0.693}{k}$$

$\therefore$ Half-life period for a first order reaction is a constant quantity at given temperature.

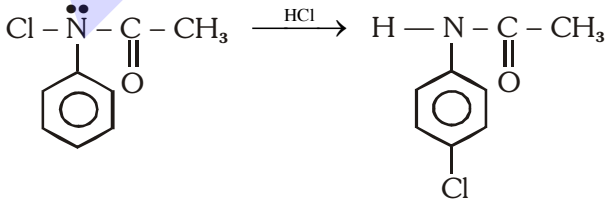
- **Examples of first order reactions :**

(i) Decomposition of azoisopropane



$$(CH_3)_2CH - N = N - CH(CH_3)_2 (g) \xrightarrow{\Delta} N_2(g) + C_6H_{14}(g)$$

(ii) Conversion of N–chloro acetanilide into p–chloroacetanilide



$$Cl - \ddot{N} - C(=O) - CH_3 \xrightarrow{HCl} H - N(Cl) - C(=O) - CH_3$$

(iii) $H_2O_2(aq.) \longrightarrow H_2O(l) + \frac{1}{2} O_2(g)$

(iv) $NH_4NO_2 \longrightarrow 2H_2O(g) + N_2(g)$

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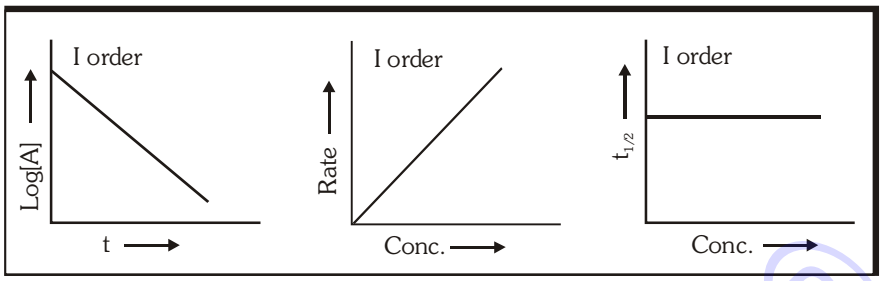
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(v) Radioactive decay

All radioactive decays always follow first order kinetics.

$^{226}_{88}Ra \longrightarrow {}^{222}_{86}Rn + {}_2He^4$

- **Graphs :**



Ex.3 **Calculate $\frac{t_{0.75}}{t_{0.50}}$ for a first order reaction :**

Sol. $k = \frac{2.303}{t_{3/4}} \log \frac{C_0}{\frac{1}{4}C_0} = \frac{2.303}{t_{1/2}} \log \frac{C_0}{\frac{C_0}{2}} \Rightarrow \frac{t_{3/4}}{t_{1/2}} = \frac{\log 4}{\log 2} = \frac{2 \log 2}{\log 2} = 2$

Ex.4 **90% of a first order reaction was completed in 10 hours. When will 99.9% of the reaction complete ?**

Sol. $K = \frac{2.303}{t} \log \frac{a}{a ? x}$

$a = 100$, $x = 90$, $t = 10$

So $K = \frac{2.303}{10} \log \frac{100}{10}$

$K = 2.303 \times 10^{-1}$ hour^{-1}

Now for 99.9% completion -

$a = 100$ and $x = 99.9$

$t = \frac{2.303}{K} \log \frac{100}{0.1} = \frac{2.303}{2.303 \times 10^{-1}} \times 3 = 30$ hours

Ex.5 **A first order reaction is 90% complete after 40 min. Calculate the half life of reaction.**

Sol. $a = 100$, $x = 90$

$K_1 = \frac{2.303}{t} \log \frac{a}{a - x}$

$= \frac{2.303}{40} \log \frac{100}{10}$

$= \frac{2.303}{40} = 5.757 \times 10^{-2}$ min^{-1}

$t_{1/2} = \frac{0.693}{K_1} = \frac{0.693}{5.757 \times 10^{-2}} = 12.03$ min.

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5.3.1 Second order reaction :

- **Case : 1**

$$\underset{a}{A} + \underset{a}{A} \longrightarrow \text{Products}$$
$$(a-x) \quad (a-x)$$

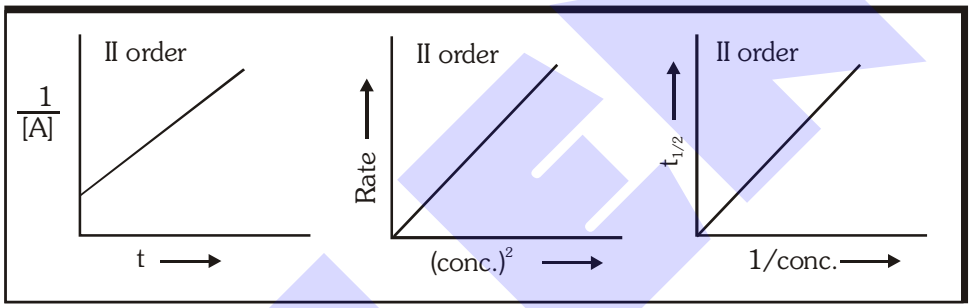
$$\therefore \quad \frac{dx}{dt} = k\,(a-x)^2$$

$$\Rightarrow \quad \frac{1}{(a-x)} - \frac{1}{a} = kt \quad \text{or} \quad \frac{1}{C_t} - \frac{1}{C_0} = kt$$

- Unit of $k = L mol^{-1} sec^{-1}$

- Half life, $t_{½} = \frac{1}{k.C_o}$

- **Graphs :**



5.3.2 Case : 2

$$A + B \longrightarrow \text{products.}$$
$$a \quad b \quad 0$$
$$a-x \quad b-x$$

Rate law

$$\frac{dx}{dt} = k\,(a-x)\,(b-x)$$

$$\int_0^x \frac{dx}{(a-x)\,(b-x)} = \int_0^t k\,dt$$

$$\boxed{k = \frac{2.303}{t(a-b)} \log \frac{b(a-x)}{a(b-x)}}$$ (no need to remember)

where $a \neq b$

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Ex.6 **For a second order reaction in which both the reactants have equal initial concentration, the time taken for 60% completion of reaction is 3000 second. What will be the time taken for 20% of the reaction?**

Sol. $k_2 = \frac{1}{t}\frac{x}{a(a-x)}$

Let a = 1,

$k_2 = \frac{1}{t}\frac{x}{(1-x)}$

$= \frac{1}{3000}\left(\frac{0.6}{1-0.6}\right)$

$= \frac{1}{3000}\times\frac{0.6}{0.4} = \frac{1}{2000}$

So time for the 20% completion :

$t = \frac{1}{k_2}\frac{x}{a(a-x)}$

$= 2000\times\frac{0.20}{0.80} = 500$ sec.

5.4 Pseudo order reaction :

For, A + B $\longrightarrow$ Products

t=0 a b

t=t a–x b–x

Rate = K [A]^n [B]^m

If [B] = constant $\Rightarrow$ [b – x] $\approx$ constant

Rate = K' [A]^n

when K' = K[B]^n

- **Case 1 :**

If concentration of B is much greater than A, then [B] = Constant $\Rightarrow$ [b – x] $\approx$ Constant.

- **Case 2 :**

If B is a catalyst , then [B] = Constant $\Rightarrow$ [b – x] $\approx$ Constant.

- **Case 3 :**

If B is a solvent , then [B] = Constant $\Rightarrow$ [b – x] $\approx$ Constant.

5.4.1 PSEUDO FIRST ORDER REACTIONS :

A second order (or of higher order) reactions can be converted into a first order reaction if the other reactant is taken in large excess. Such first order reactions are known as **pseudo first order reactions**.

$\therefore$ For A + B $\longrightarrow$ Products **[Rate] = K [A]^1 [B]^1**

$K = \frac{2.303}{t\ (a-b)}\log\frac{b(a-x)}{a(b-x)}$ (from 5.3.2)

Now if 'B' is taken in large excess b > > a.

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$$\therefore \quad K = \frac{2.303}{-bt}\log\frac{(a-x)}{a} \Rightarrow K = \frac{2.303}{bt}\log\frac{a}{a-x}$$

$$\Rightarrow \quad K.b = \frac{2.303}{t}\log\frac{a}{a-x} \Rightarrow K' = \frac{2.303}{t}\log\frac{a}{a-x}$$

K' is pseudo first order rate constant.

K' will have units of first order.

K will have units of second order.

❖ **Table :** Characteristics of Zero, First, Second and n^{th} order reactions of the type **A ⟶ Products**

	Zero order	**First order**	**Second order**	**n^{th} order**
Differential Rate law	$-\frac{d[A]}{dt} = k[A]^\circ$	$-\frac{d[A]}{dt} = k[A]$	$-\frac{d[A]}{dt} = k[A]^2$	$-\frac{d[A]}{dt} = k[A]^n$
Integrated Rate law	$[A]_t = [A]_0 - kt$	$\ln[A]_t = -kt + \ln[A]_0$	$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$	$\frac{1}{[A_t]^{n-1}} - \frac{1}{[A_0]^{n-1}} = (n-1)\ kt$
Linear graph	$[A]_t$ v/s t	ln [A] v/s t	$\frac{1}{[A]}$ v/s t	$\frac{1}{[A_t]^{n-1}}$ v/s t
Half-life	$t_{1/2} = \frac{[A]_0}{2k}$ (depends on $[A]_0$)	$t_{1/2} = \frac{0.693}{k}$ (independent of $[A]_0$)	$t_{1/2} = \frac{1}{k[A]_0}$ (depends on $[A]_0$)	$t_{1/2} \propto \frac{1}{[A_0]^{n-1}}$

6 Methods to determine order of a reaction :

6.1 Initial rate method :

By comparison of different initial rates of a reaction by varying the concentration of one of the reactants while others are kept constant.

$r = k[A]^a[B]^b[C]^c$ if [B] = constant and [C] = constant

then for two different initial concentrations of A we have

$$r_{0_1} = k[A_0]_1^a \quad r_{0_2} = k[A_0]_2^a \Rightarrow \frac{r_{0_1}}{r_{0_2}} = \left(\frac{[A_0]_1}{[A_0]_2}\right)^a$$

or in log form we have $$a = \frac{\log\left(r_{0_1}/r_{0_2}\right)}{\log\left([A_0]_1/[A_0]_2\right)}$$

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6.2 Integrated rate law method :

It is method of hit and trial. By checking if the kinetic data (experimental data) best fits into a particular integrated rate law, we determine the order. It can also be done graphically.

Ex.7 The rate of decomposition of N_2O_5 in CCl_4 solution has been studied at 318 K and the following results have been obtained :

t/min	0	135	342	683	1693
c/M	2.08	1.91	1.67	1.35	0.57

Find the order of the reaction and calculate its rate constant. What is the half-life period?

Sol. It can be shown that these data will not satisfy the integrated rate law of zero order. We now try integrated first order equation i.e., $k = \frac{\ln(c_0/c)}{t}$.

t / min	c / M	$k = \frac{\ln(c_0/c)}{t}$ min^{-1}
0	2.08	6.32×10^{-4}
135	1.91	6.30×10^{-4}
339	1.68	6.32×10^{-4}
683	1.35	6.32×10^{-4}
1680	0.72	6.31×10^{-4}

It can be seen that the value of k is almost constant for all the experimental results and hence it is first order reaction with $k = 6.31 \times 10^{-4}$ min^{-1}.

$$t_{1/2} = \frac{0.693}{6.31 \times 10^{-4}\ \text{min}^{-1}} = 1.094 \times 10^{3}\ \text{min}^{-1}$$

6.3 Method of half lives :

The half lives of each order is unique so by comparing half lives we can determine order for n^{th} order reaction $t_{1/2} \propto \frac{1}{[A_0]^{n-1}}$ (Remember)

$$\frac{(t_{1/2})_1}{(t'_{1/2})_2} = \frac{[A'_0]_2^{\,n-1}}{[A_0]_1^{\,n-1}}$$ (Remember)

Ex.8 In a decomposition reaction, the reaction was found to be 50% complete in 210 seconds when the initial pressure of the mixture was 200 mm. In a second experiment the time of half reaction was 140 seconds when the initial pressure was 300 mm. Calculate the total order of the reaction.

Sol. For a nth order reaction $t_{1/2} \propto \frac{1}{c_0^{n-1}}$

$$\frac{210}{140} = \left(\frac{300}{200}\right)^{n-1} \Rightarrow n = 2$$

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6.4 Ostwald's isolation method :

This method is useful for reactions which involve a large number of reactants. In this method, the concentration of all the reactants are taken in large excess except that of one, so if

$$\textbf{Rate} = k[A]^a[B]^b[C]^c = k_0[A]^a$$

Then value of 'a' can be calculated by previous methods and similarly 'b' and 'c' can also be calculated.

6.5 Initial rate method :

For reaction : A + B ⟶ Products

Initial rate, $r_0 = K.[A_0]^n [B_0]^m$

Now, order with respect to A may be determined by comparing the initial rate of reaction at different initial concentration of A but fixed initial concentration of B.

Ex.9 Consider the following data for the reaction :

A + B ⟶ Products

Run	Initial concentration	Initial concentration	Initial rate (mol s^{-1})
1	0.10 M	1.0 M	2.1×10^{-3}
2.	0.20 M	1.0 M	8.4×10^{-3}
3.	0.20 M	2.0 M	8.4×10^{-3}

Determine the order of reaction with respect to A and with respect B and the over all order of reaction.

Sol. The rate law may be expressed as :

Rate = $k[A]^p[B]^q$

Comparing experiments 2 and 3

$(Rate)_2 = k[0.2]^p[1.0]^q = 8.4 \times 10^{-3}$ (1)

$(Rate)_3 = k[0.2]^p[2.0]^q = 8.4 \times 10^{-3}$ (2)

Dividing equation (2) by (1)

$$\frac{(Rate)_3}{(Rate)_2} = \frac{k[0.2]^p[2.0]^q}{k[0.2]^p[1.0]^q} = \frac{8.4\times10^{-3}}{8.4\times10^{-3}}$$

$[2]^q = [2]^0$

or $q = 0$

Comparing experiments 1 and 2,

$(Rate)_2 = k[0.20]^p[1.0]^q = 8.4 \times 10^{-3}$ (3)

$(Rate)_1 = k[0.10]^p[1.0]^q = 2.1 \times 10^{-3}$ (4)

Dividing equation (3) by (4)

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$$\frac{(\text{Rate})_2}{(\text{Rate})_1} = \frac{k[0.20]^p[1.0]^q}{k[0.10]^p[1.0]^q} = \frac{8.4\times10^{-3}}{2.1\times10^{-3}} = 4$$

$[2]^p = [2]^2$ or $p = 2$

so order with respect to A = 2

order with respect to B = 0

overall order = 2

7. APPLICATION OF FIRST ORDER REACTION

(Methods to monitor the progress of the reaction)

7.1 Case : 1

First order gaseous reaction :

Progress of gaseous reactions can be monitored by measuring total pressure at a fixed volume & temperature. This method can be applied for those reactions also in which a gas is produced because of decomposition of a solid or liquid. We can get an idea about the concentration of reacting species at a particular time by measuring pressure.

The pressure data can be given in terms of

(i) Partial pressure of the reactant

(ii) Total pressure of the reaction system

(iii) Pressure at only some points of time.

Ex.10 **Find the expression for K in terms of P_0 , P_t and n. For the reaction**

$$\mathbf{A(g) \longrightarrow nB(g)}$$

Sol.

$$\mathbf{A(g) \longrightarrow nB(g)}$$

P_0

$P_A = (P_0 - x)$ nx

$\therefore$ P_t (Total pressure at time 't') $= P_0 - x + nx = P_0 + (n-1)\,x$

$\therefore$ $x = \dfrac{P_t - P_0}{n-1}$

$\therefore$ $P_A = P_0 - \dfrac{P_t - P_0}{n-1} = \dfrac{P_0 n - P_t}{n-1}$

$\therefore$ $a \propto p_0$ & $(a-x) \propto P_A = \dfrac{nP_0 - P_t}{n-1}$

$\therefore$ $k = \dfrac{2.303}{t} \log \dfrac{P_0(n-1)}{(nP_0 - P_t)}$

Final total pressure after infinite time $= P_f = nP_0$

Do not remember the formula but derive it for each question.

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7.2 Case : 2

Volume measurement :

(i) By measuring the volume of product formed we can monitor the progress of reactions.

Ex.11 Study of a reaction whose progress is monitored by measuring the volume of a escaping gas.

$$NH_4NO_2(s) \xrightarrow{\Delta} 2H_2O(\ell) + N_2(g)$$

Sol. Let, V_t be the volume of N_2 collected at time 't'

V_∞ = be the volume of N_2 collected at the end of the reaction.

$$a \propto V_\infty \text{ and } \quad x \propto V_t$$

$$(a - x) \propto V_\infty - V_t$$

$$\therefore \quad k = \frac{2.303}{t} \log \frac{V_\infty}{V_\infty - V_t}$$

(ii) By titration method : By measuring the volume of titrating agent we can monitor amount of reactant remaining or amount of product formed at any time. It is the **titre value** . Here the milliequivalent or millimoles are calculated using valence factors.

Ex.12 From the following data show that the decomposition of hydrogen peroxide in aqueous solution is a first - order reaction. What is the value of the rate constant?

Time in minutes	0	10	20	30	40
Volume V in mL	25.0	20.0	15.7	12.5	9.6

Where V is the volume in mL of standard $KMnO_4$ solution required to react with a definite volume of hydrogen peroxide solution.

Sol. The equation for a first order reaction is,

$$2H_2O_2(\ell) \xrightarrow{\Delta} 2H_2O(\ell) + O_2(g)$$

the volume of $KMnO_4$ used, evidently corresponds to the undecomposed hydrogen peroxide.

Hence the volume of $KMnO_4$ used, at zero time corresponds to the initial concentration a and the volume used after time t, corresponds to (a – x) at that time. Inserting these values in the above equation, we get

When t = 10 min. $k_1 = \frac{2.303}{10} \log \frac{25}{20.0} = 0.022318 \text{ min}^{-1} = 0.000372 \text{ s}^{-1}$

When t = 20 min. $k_1 = \frac{2.303}{20} \log \frac{25}{15.7} = 0.023265 \text{ min}^{-1} = 0.000387 \text{ s}^{-1}$

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When $t = 30$ min. $k_1 = \frac{2.303}{30} \log \frac{25}{12.5} = 0.02311 \text{ min}^{-1} = 0.000385 \text{ s}^{-1}$

When $t = 40$ min. $k_1 = \frac{2.303}{40} \log \frac{25}{9.6} = 0.023932 \text{ min}^{-1} = 0.0003983 \text{ s}^{-1}$

The constancy of k, shows that the decomposition of H_2O_2 in aqueous solution is a **first order** reaction. The average value of the rate constant is 0.0003879 s^{-1}.

Ex.13 Study of acid hydrolysis of an ester.

$$CH_3COOCH_3 + \underset{(\text{excess})}{H_2O} \xrightarrow[(HCl)]{H^+} CH_3COOH + CH_3OH$$

The progress of this reaction is monitored or determined by titrating the fixed volume of reaction mixture at different time intervals against a standard solution of NaOH using phenolphthalein as indicator. Find out rate constant of the reaction in terms of volume of NaOH consumed at $t = 0$, V_0, at $t = \infty$, V_∞ & at time t, V_t.

Sol. Let, V_0 = Volume of NaOH used at t = 0 [this is exclusively for HCl.]

V_t = Volume of NaOH used at 't'

V_∞ = Volume of NaOH used at $t = \infty$

$a \propto (V_\infty - V_0)$

$(a - x) \propto (V_\infty - V_t)$

$x \propto (V_t - V_0)$

$$k = \frac{2.303}{t} \log \frac{V_\infty - V_0}{V_\infty - V_t}$$

7.3 Case : 3

Optical rotation measurement :

It is used for optically active sample. It is applicable if there is at least one optically active species involved in chemical reaction.

- The optically active species may be present in reactant or product.

It is found that $(r_\infty - r_0) \propto a$ (a = concentration , x = amount consumed)

$(r_\infty - r_t) \propto (a - x)$

Where are r_0, r_t, r_∞ are angle of optical rotation at time $t = 0$, $t = t$ and $t = \infty$.

$$k = \frac{2.303}{t} \log \frac{r_\infty - r_0}{r_\infty - r_t}$$ (Remember)

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Ex.14 **Study of hydrolysis of sucrose, progress of this reaction is monitored with the help of polarimeter because a solution of sucrose is dextrorotatory and on hydrolysis, the mixture of glucose and fructose obtained becomes laevorotatory. That's why this reaction is also known as inversion of cane sugar.**

$$\underset{\text{excess}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} \underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{fructose}}{C_6H_{12}O_6}$$

Let the readings in the polarimeter be

$t = 0$, θ_0 ; $t =$ 't', θ_t and at $t = \infty$, θ_∞

Then calculate rate constant 'k' in terms of these readings.

Sol. The principle of the experiment is that change in the rotation is directly proportional to the amount of sugar hydrolysed.

$\therefore \quad a \propto (\theta_0 - \theta_\infty) \quad ; \quad (a - x) \propto (\theta_t - \theta_\infty) \; ; \; x \propto (\theta_0 - \theta_t)$

$$k = \frac{2.303}{t} \log\left(\frac{\theta_0 - \theta_\infty}{\theta_t - \theta_\infty}\right)$$

7.4 Case : 4

First order growth reaction :

For bacteria multiplication or virus growth use following concept

Consider a growth reaction,

Time	**Population (or colony)**
0	a
t	(a + x)

$$\frac{dx}{dt} = k(a + x) \quad \text{or} \quad \frac{dx}{(a+x)} = kdt$$

On integration

$\log_e (a + x) = kt + C \quad$ at $\quad t = 0 \; ; \; x = 0 \Rightarrow C = \log_e a$

$$kt = -\log_e \frac{a}{(a+x)} = -\frac{2.303}{t} \log_{10}\left(\frac{a}{(a+x)}\right)$$

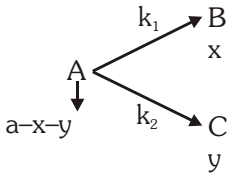
$$\boxed{k = \frac{2.303}{t} \log_{10}\left(\frac{a+x}{a}\right)}$$

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❖ **Generation time :** At t = generation time , $x = a$ $\therefore$ $t = \frac{0.693}{K}$

8. SOME SPECIAL CASES :

8.1 FIRST ORDER PARALLEL OR COMPETING OR SIMULTANEOUS REACTIONS



At $t = 0$ $[A] = a$ $[B] = [C] = 0$

(i) Differential rate law :

$$\frac{d[B]}{dt} = k_1[A] \; ; \; \frac{d[C]}{dt} = k_2[A]$$

and, $$\frac{-d[A]}{dt} = \frac{d[B]}{dt} + \frac{d[C]}{dt} \Rightarrow \frac{-d[A]}{dt} = (k_1 + k_2)[A] = k_{eff}[A]$$

k_{eff} = $k_1 + k_2$ = overall rate constant for the disappearance of 'A'

(ii) Integral rate law :

$$[A]_t = ae^{-k_{eff}t} = ae^{-(k_1+k_2)t}$$

$$\frac{d[B]}{dt} = k_1[A] \Rightarrow \frac{d[B]}{dt} = k_1 a e^{-(k_1+k_2)t}$$

$$[B] = \left(\frac{k_1 a}{k_1 + k_2}\right)(1 - e^{-(k_1+k_2)t})$$

Similarly, $$[C] = \left[\frac{k_2 a}{k_1 + k_2}\right](1 - e^{-(k_1+k_2)t})$$

(iii) Composition of product :

$$\frac{[B]}{[C]} = \frac{k_1}{k_2}$$

8.2 FIRST ORDER REVERSIBLE REACTION

$$A \underset{k_b}{\overset{k_f}{\rightleftharpoons}} B$$

$t = 0$	a	0
$t = t$	$a - x$	x
$t = t_{eq.}$	$a - x_{eq.}$	$x_{eq.}$

$x_{eq.}$ = eq conc. of product

$$-\frac{d[A]}{dt} = +k_f[A] - k_b[B]$$

At equilibrium $\frac{d[A]}{dt} = 0, \; k_f[A] = k[B]$

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9. EFFECT OF TEMPERATURE ON RATE OF REACTION :

In early days the effect of temperature on reaction rate was expressed in terms of ***temperature coefficient*** (μ) which was defined as the ratio of rate of reaction at two different temperature differing by 10ºC (usually these temperatures were taken as 25ºC and 35ºC)

$$\text{T.C, } \mu = \frac{K_{T+10}}{K_T} \approx 2 \text{ to } 3 \quad \textbf{(for most of the reactions)}$$

Note : Rate of reaction increases on increasing temperature (Generally ; T.C > 1)

Ex.15 **For a reaction T.C. = 2, Calculate $\frac{k_{40°C}}{k_{25°C}}$ for this reaction. Assuming T.C remains constant.**

Sol. $\frac{k_2}{k_1} = (\text{T.C.})^{\frac{\Delta t}{10}} = (2)^{\frac{15}{10}} = (2)^{\frac{3}{2}} = \sqrt{8}$

But the method of temperature coefficient was not exact to explain the effect of temperature on reaction rate. For that a new theory was evolved.

9.1 Collision theory of reaction rate :

It was developed by ***Max Trautz*** and ***William lewis.*** It gives insight in to the energetics and mechanistic aspects of reactions.

It is based upon kinetic theory of gases.

According the this theory :

(i) A chemical reaction takes place due to the collision among reactant molecules. The number of collisions taking place per second per unit volume of the reaction mixture is known as collision frequency (Z).

(ii) Every collision does not bring a chemical change. The collision that actually produce the products are effective collision. For a collision to be effective the following two barriers are to be cleared.

❖ **Energy barrier :** The minimum amount of absolute energy which the colliding molecules must possess as to make the chemical reaction to occur is known as **threshold energy**.

- Reactant molecules having energy equal or greater than the threshold are called ***active molecules*** and those having energy less than the threshold are called ***passive molecules***.
- At a given temperature there exists a dynamic equilibrium between active and passive molecules. The process of transformation from passive to active molecules being endothermic, increase of temperature increases the number of active molecules and hence the reaction.

Passive molecules $\rightleftharpoons$ Active molecules, ΔH = +ve

- "The minimum amount of excess energy required by reactant molecules to participate in a reaction is called **activation energy (E_a)".**

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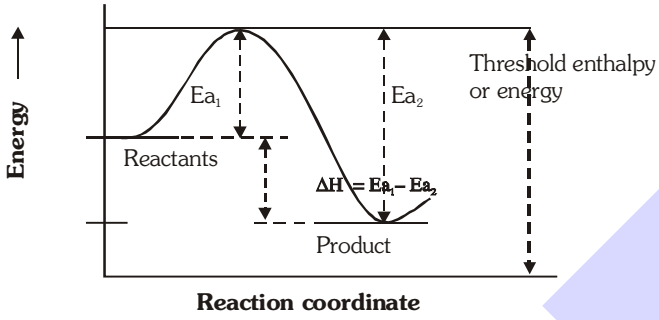
CONCEPT OF ENERGY OF ACTIVATION (E_a) :

- The average extra amount of energy which the reactant molecules (having energy less than the threshold) must acquire so that their mutual collision may lead to the breaking of bond(s) and hence the energy is known as energy of activation of the reaction. It is denoted by the symbol E_a. Thus,

  E_a = Threshold energy – Actual average energy of reactants

  E_a is expressed in kcal mole^{-1} or kJ mole^{-1}.

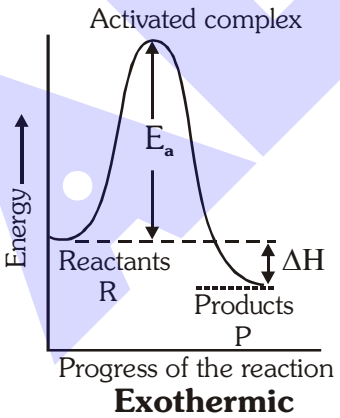
- The essence of Collision Theory of reaction rate is that there exists an energy barrier in the reaction path between reactant(s) and product(s) and for reaction to occur, the reactant molecules must climb over the top of the barrier which they do by collision. The existence of energy barrier and concept of E_a can be understood from the following diagram.



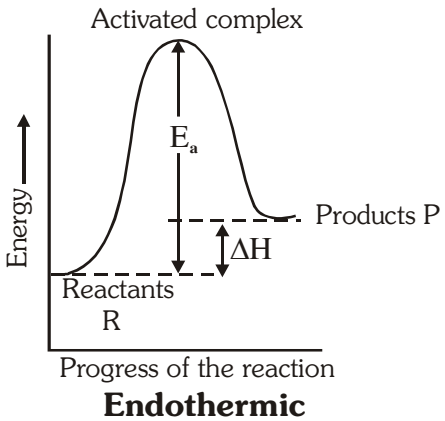
ΔH = Enthalpy change during the reaction (R → P)
Ea_1 = Energy of activation of the forward reaction
Ea_2 = Energy of activation of the backward reaction

From the figure above it can be concluded that the minimum activation energy of any exothermic reaction will be zero while minimum activation energy for any endothermic reaction will be equal to ΔH.

Greater the height of energy barrier, greater will be the energy of activation and more slower will be the reaction at a given temperature.



Exothermic

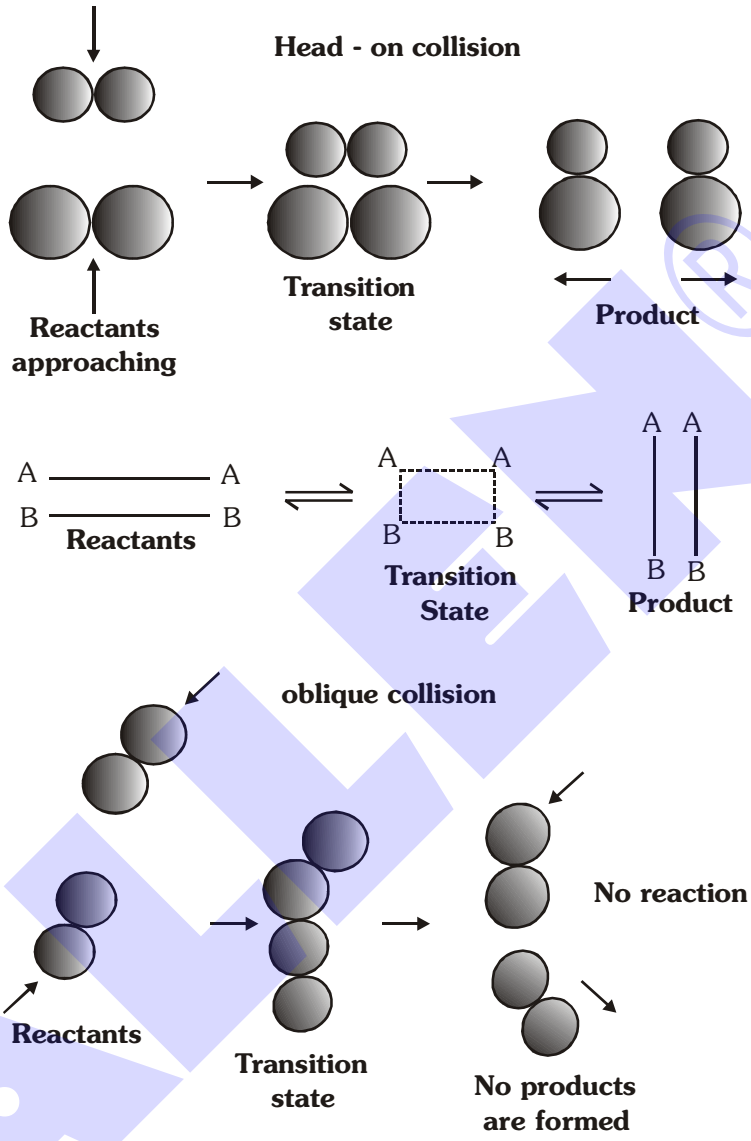


Endothermic

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❖ **Orientation barrier :** Energy alone does not determine the effectiveness of the collision. The reacting molecules must collide in proper direction to make collision effective. Following diagrams can explain importance of suitable direction for collision.

Consider reaction : $A_2 + B_2 \rightarrow 2AB$

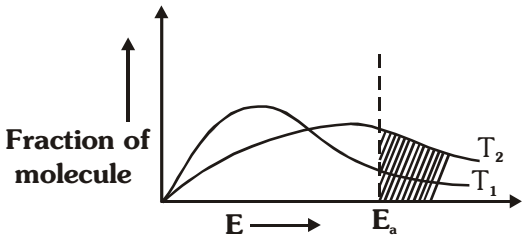


(iii) **Rate of any chemical reaction** = Collision frequency × fraction of the total number of effective collision.

= Collision frequency × fraction of the total number of collision in which K.E. of the colliding molecules equals to E_a or exceeds over it × fraction of collision in proper orientation.

From Maxwellian distribution, it is found that fraction of molecules having excess energy greater than threshold energy lead to the formation of product.

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$T_2 > T_1$

$e^{-E_a/RT} \rightarrow$ Represents fraction of molecules having K.E. greater than or equal to E_a.

$$\text{Rate} \propto e^{-E_a/RT}$$

Dependence of rate on temperature is due to dependence of k on temperature.

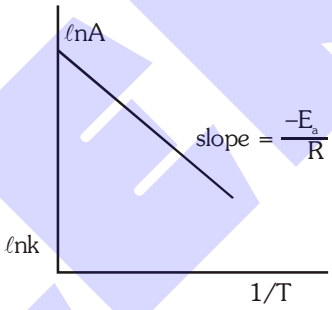
$$k \propto e^{-E_a/RT}$$

$$\mathbf{k = Ae^{-E_a/RT}} \qquad \textbf{[Arrhenius equation]}$$

'A' is **pre-exponential** factor or **frequency factor** representing collisions taking place with proper orientation. A and E_a are assumed to be independent of temperature.

$$\ell nk = \ell n\, A - \frac{E_a}{RT}$$

As $T \rightarrow \infty$, $K \rightarrow A$



At temperature T_1, Rate constant = k_1
At temperature T_2, Rate constant = k_2

$$\ell nk_1 = \ell nA - \frac{E_a}{RT_1} \;\&\; \ell nk_2 = \ell nA - \frac{E_a}{RT_2} \Rightarrow \ell n\,\frac{\mathbf{k_2}}{\mathbf{k_1}} = \frac{\mathbf{E_a}}{\mathbf{R}}\left(\frac{1}{\mathbf{T_1}} - \frac{1}{\mathbf{T_2}}\right)$$

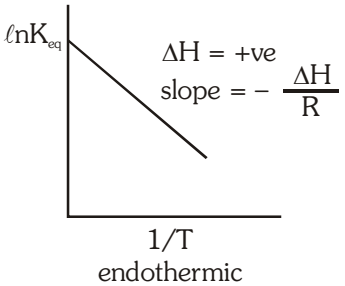
9.2 REVERSIBLE REACTIONS

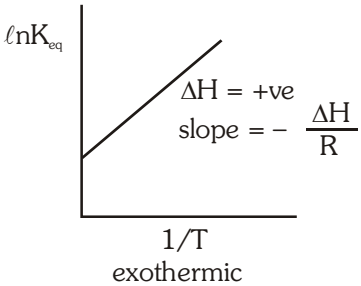
$$k_f = A_f\, e^{-E_{af}/RT}$$

$$k_b = A_b\, e^{-E_{ab}/RT}$$

$$K_{eq} = \frac{k_f}{k_b} = \frac{A_f\, e^{-E_{af}/RT}}{A_b\, e^{-E_{ab}/RT}} = \left(\frac{A_f}{A_b}\right) e^{-(E_{af}-E_{ab})/RT}$$

$$\ell n\, K_{eq} = -\frac{\Delta H}{RT} + \ell n\left(\frac{A_f}{A_b}\right)$$





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Ex.16 **For a reaction, temperature coefficient = 2, then calculate the activation energy (in kJ) of the reaction.**

Sol. **Given :** Temperature coefficient $= \frac{K_2}{K_1} = 2$

$T_1 = 25 + 273 = 298$ k

$T_2 = 35 + 273 = 308$ k

$$\log\frac{K_2}{K_1} = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1T_2}\right)$$

$$\log 2 = \frac{E_a}{2.303\times 8.314}\times\left(\frac{10}{298\times 308}\right)$$

$E_a = 52.31$ kJ mol^{-1}

Ex.17 **For first order gaseous reaction, log k when plotted against $\frac{1}{T}$, it give a straight the with a slope of –8000. Calculate the activation energy of the reaction.**

Sol. For an arrhenius equation $k = Ae^{-E_a/RT}$

$$\log k = \log A - \frac{E_a}{2.303R}\times\frac{1}{T}$$

when curve is plotted between log k and $\frac{1}{T}$. A straight line is obtained and a slope of this curve

$$= -\frac{E_a}{2.303R}$$

Then, $\frac{E_a}{2.303R} = 8000$

or $E_a = 8000 \times 2.303 \times 2 = 36848$ calories

Ex.18 **The slope of the plot of log k vs $\frac{1}{T}$ for a certain reaction was found to be – 5.4 × 10^3. Calculate the energy of activation of the reaction. If the rate constant of the reaction is 1.155 × 10^{-2} sec^{-1} at 373 K , what is its frequency factor ?**

Sol. **(a)** $\text{Slope} = \frac{-E}{2.303\ \ R} = -5.4\times 10^3$

$E_a = 5.4 \times 10^3 \times 2.303 \times 1.987$

$= 24.624$ cal mol^{-1}

(b) $K = Ae^{-E/RT}$; $\log 1.155\times 10^{-2} = \log A - \frac{24.624}{2.303\times 1.987\times 373}$

or $A = 1.764 \times 10^3$ sec^{-1}

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10. CATALYST AND CATALYSIS :

A catalyst is a substance, which increases the rate of a reaction without itself being consumed at the end of the reaction, and the phenomenon is called ***catalysis***.

Thermal decomposition of $KClO_3$ is found to be accelerated by the presence of MnO_2. Here MnO_2 acts as a catalyst.

$$2KClO_3 + [MnO_2] \longrightarrow 2KCl + 3O_2 \uparrow + [MnO_2]$$

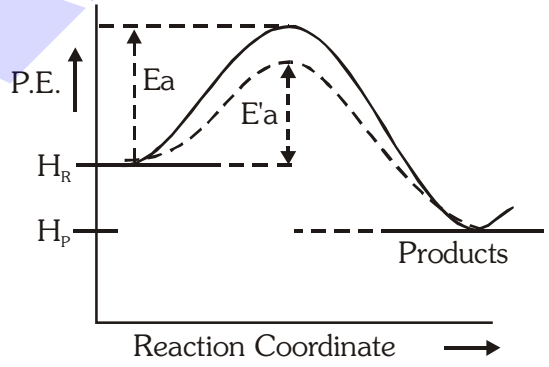
- MnO_2 can be received in the same composition and mass at the end of the reaction.
- The word catalyst should not be used when the added substance reduces the rate of reaction. The substance is then called **inhibitor**.
- Catalyst are generally foreign substances but sometimes one of the product may act as a catalyst and such catalyst is called ***"auto catalyst"*** and the phenomena is called ***auto catalysis***.

In the permanganate titration of oxalic acid initially there is slow discharge of the colour of permanganate solution but afterwards the discharge of the colour become faster. This is due to the formation of $MnSO_4$ during the reaction which acts as a catalyst for the same reaction. Thus, $MnSO_4$ is an ***"auto catalyst"*** for this reaction. This is an example of auto catalyst.

$$2KMnO_4 + 3H_2SO_4 + 5H_2C_2O_2 \longrightarrow K_2SO_4 + 8H_2O + 10CO_2 + MnSO_4$$

10.1 General characteristics of catalyst :

- A catalyst does not initiate the reaction normally. It simply fastens it.
- Only a small amount of catalyst can catalyse the reaction.
- A catalyst does not alter the position of equilibrium i.e. magnitude of equilibrium constant and hence ΔG°. It simply lowers the time needed to attain equilibrium. This means if a reversible reaction in absence of catalyst completes to go to the extent of 75% till attainment of equilibrium, and this state of equilibrium is attained in 20 minutes then in presence of a catalyst, the reaction will still go to 75% of completion on the attainment of equilibrium but the time needed for this will be less than 20 minutes.



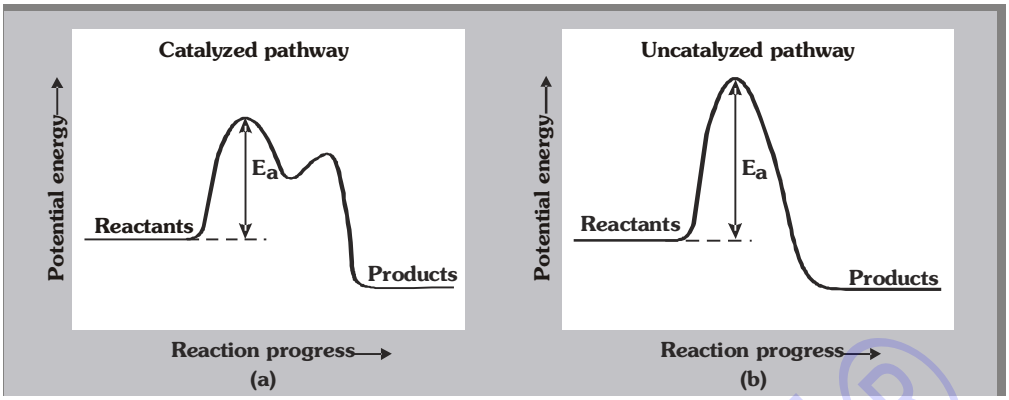
A catalyst drives the reaction through a low energy path and hence E_a is less. That is, the function of the catalyst is to lower down the activation energy.

E_a = Energy of activation in absence of catalyst.

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E_a' = Overall Energy of activation in presence of catalyst.

$E_a - E_a'$ = Lowering of activation energy by catalyst.



10.2 Comparison of rates of reaction in presence and absence of catalyst :

If k and k_{cat} be the rate constant of a reaction at a given temperature T , E_a and E'_a are the activation energies of the reaction in absence and presence of catalyst, respectively, the

$$\frac{k_{cat}}{k} = \frac{Ae^{-E'a/RT}}{Ae^{-Ea/RT}} = Ae^{(E_a - E'_a)/RT}$$

Since $E_a - E'_a$ is positive so $k_{cat} > k$. the ratio $\frac{k_{cat}}{k}$ gives the number of times the rate of reaction will increase by the use of catalyst at a given temperature.

The rate of reaction in the presence of catalyst at any temperature T_1 may be made equal to the rate of reaction in absence of catalyst but for this sake we will have to raise the temperature. Let this temperature

be $T_2 = e^{-E'_a/RT_1} = e^{-E_a/RT_2}$ or $\frac{E'_a}{T_1} = \frac{E_a}{T_2}$

11 CALCULATION OF RATE LAW / ORDER

11.1 When first step is rate determining step.

Ex.19 Calculate order and rate law of reaction -

$$2NO_2 + F_2 \longrightarrow 2NO_2F$$

with help of mechanism

I. $NO_2 + F_2 \xrightarrow{K_1} NO_2F + F$ **(slow)**

II. $NO_2 + F \xrightarrow{K_2} NO_2F$ **(fast)**

Sol. According to RDS

Rate = $k_1 [NO_2] [F_2]$

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11.2 Equilibrium approach :

Ex.20 **For the reaction , $H^+ + HNO_2 + C_6H_5NH_2 \xrightarrow{Br^-} C_6H_5N_2^+ + 2H_2O$, the mechanism is ,**

$H^+ + HNO_2 \underset{k_2}{\overset{k_1}{\rightleftharpoons}} H_2NO_2^+$ (fast equilibrium step)

intermediate

$H_2NO_2^+ + Br^- \xrightarrow{k_3} NOBr + H_2O$ (slow)

$NOBr^- + C_6H_5NH_2 \xrightarrow{k_4} C_6H_5N_2^+ + Br^- + H_2O$ (fast)

Derive the rate law expression for the reaction

Sol. $r = k_3 [Br^-] [H_2NO_2^+]$

$$K_{eq} = \frac{k_1}{k_2} = \frac{[H_2NO_2^+]}{[H^+][HNO_2]}$$

$$[H_2NO_2^+] = \left[\frac{k_1}{k_2}\right][H^+] [HNO_2]$$

$$r = \frac{k_1 k_3}{k_2} [H^+] [HNO_2] [Br^-].$$

Note : Rate law can have reactant product or catalyst concentration terms but not any intermediate terms.

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MISCELLANEOUS PREVIOUS YEARS QUESTION

1. Which of the following statement(s) is (are) correct **[JEE 1999]**

(A) A plot of log K_p versus 1/T is linear

(B) A plot of log [X] versus time is linear for a first order reaction, $X \longrightarrow P$

(C) A plot of log P versus 1/T is linear at constant volume.

(D) A plot of P versus 1/V is linear at constant temperature.

Ans. **(A,B,D)**

2. The rate constant for an isomerisation reaction $A \rightarrow B$ is 4.5×10^{-3} min^{-1}. If the initial concentration of A is 1 M. Calculate the rate of the reaction after 1 h. **[JEE 1999]**

Ans. **3.435×10^{-3} M/min**

Sol. $r = k[A]$

$r = K[A]_0 e^{-kt}$

$r = (4.5 \times 10^{-3}) e^{-4.5 \times 10^{-3} \times 60}$ M/min.

3. A hydrogenation reaction is carried out at 500 K. If the same reaction is carried out in the presence of a catalyst at the same rate, the temperature required is 400 K. Calculate the activation energy of the reaction if the catalyst lowers the activation barrier by 20 $kJmol^{-1}$. **[JEE 2000]**

Ans. **100 $kJmol^{-1}$**

Sol. $$\ln\left(\frac{r_2}{r_2}\right) = -\frac{1}{R}\left(\frac{Ea_2}{T_2} - \frac{Ea_1}{T_1}\right)$$

$$\ln(1) = -\frac{1}{R}\left(\frac{Ea_1 - 20}{40} - \frac{Ea_1}{500}\right)$$

$Ea_1 = 100$

4. The rate constant for the reaction **[JEE SCR 2000]**

$$2N_2O_5 \longrightarrow 4NO_2 + O_2$$

is 3.0×10^{-5} sec^{-1}. if the rate is 2.4×10^{-5} mol $litre^{-1}$ sec^{-1}, then the concentration of N_2O_5 (in mol $litre^{-1}$) is

(A) 1.4 (B) 1.2 (C) 0.004 (D) 0.8

Ans. **(D)**

Sol. $r = k[N_2O_5]^2$

$2.4 \times 10^{-5} = 3 \times 10^{-5}[N_2O_5]^2$

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5. If I is the intensity of absorbed light and C is the concentration of AB for the photochemical process : $AB + h\nu \longrightarrow AB$, the rate of formation of AB is directly proportional to **[JEE SCR 2001]**

(A) C (B) I (C) I^2 (D) CI

Ans. **(B)**

Sol. For phot chemical reaction

$r \propto I$

6. The rate of a first order reaction is 0.04 mole $litre^{-1}$ s^{-1} at 10 minutes and 0.03 mol $litre^{-1}$ s^{-1} at 20 minutes after initiation. Find the half life of the reaction. **[JEE 2001]**

Ans. $\mathbf{t_{1/2} = 24.14\ min}$

Sol. $\left(\frac{\ln 2}{t_{1/2}}\right)(20 - 10) = \ln\left(\frac{0.04}{0.03}\right)$

7. Consider the chemical reaction, $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$. The rate of this reaction can be expressed in term of time derivative of concentration of $N_2(g)$, $H_2(g)$ or NH_3 (g). Identify the correct relationship amongst the rate expressions. **[JEE SCR 2002]**

(A) Rate $= -d[N_2]/dt = -1/3\ d[H_2]/dt = 1/2d[NH_3]/dt$

(B) Rate $= -d[N_2]/dt = -3\ d[H_2]/dt = 2d[NH_3]/dt$

(C) Rate $= d[N_2]/dt = 1/3\ d[H_2]/dt = 1/2d[NH_3]/dt$

(D) Rate $= -d[N_2]/dt = -d[H_2]/dt = d[NH_3]/dt$

Ans. **(A)**

Sol. $\frac{-d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$

8. In a first order reaction the concentration of reactant decreases from 800 mol/dm^3 to 50 mol/dm^3 in 2×10^4 sec. The rate constant of reaction in sec^{-1} is **[JEE SCR 2003]**

(A) 2×10^4 (B) 3.45×10^{-5}

(C) 1.3486×10^{-4} (D) 2×10^{-4}

Ans. **(C)**

Sol. $K(2 \times 10^4) = \ln\left(\frac{800}{50}\right)$

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9. The reaction, X⟶ Product follows first order kinetics. In 40 minutes the concentration of X changes from 0.1 M to 0.025 M. Then the rate of reaction when concentration of X is 0.01 M

(A) 1.73×10^{-4} M min $^{-1}$ (B) 3.47×10^{-5} M min $^{-1}$

(C) 3.47×10^{-4} M min^{-1} (D) 1.73×10^{-5} M min^{-1} **[JEE SCR 2004]**

Ans. **(C)**

Sol. $K(40) = \ln\left(\frac{0.1}{0.25}\right)$

$K = \frac{\ln 2}{20}$

$r = K[X]$

$= \frac{\ln 2}{20} \times 0.1$

10. $2X(g) \longrightarrow 3Y(g) + 2Z(g)$

Time (in Min)	0	100	200
Partial pressure of X (in mm of Hg)	800	400	200

Assuming ideal gas condition. Calculate

(a) Order of reaction

(b) Rate constant (K_X)

(c) Time taken for 75% completion of reaction

(d) Total pressure when $P_X = 700$ mm. **[JEE 2005]**

Ans. **(a) 1, (b) 6.93×10^{-3} min^{-1} , (c) 200, (d) 950 mm**

Sol. (a) As half life is constant so it will be of first order.

(b) $K_x(100) = \ln\left(\frac{800}{400}\right)$

(c) $t = 75\% = 2t_{50\%} = 2\left(\frac{\ln 2}{K_x}\right)$

(d) $2X(g) \longrightarrow 3Y(g) + 2Z(g)$

$800 - x \quad \frac{3x}{2} \quad x$

Here $x = 100$

So $P_T = 800 + \frac{3x}{2} = 950$

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11. Which of the following statement is incorrect about order of reaction? **[JEE 2005]**

(A) Order of reaction is determined experimentally

(B) It is the sum of power of concentration terms in the rate law expression

(C) It does not necessarily depend on stoichiometric coefficients

(D) Order of the reaction can not have fractional value.

Ans. **(D)**

12. Consider a reaction aG + bH → Products. When concentration of both the reactants G and H is doubled, the rate increases by eight times. However, when concentration of G is doubled keeping the concentration of H fixed, the rate is doubled. The overall order of the reaction is : **[JEE 2006]**

(A) 0 (B) 1 (C) 2 (D) 3

12. **Ans.(D)**

Sol. $\frac{r_2}{r_1}\left(\frac{[G]_2}{[G]_1}\right)^{\alpha}\left(\frac{[H]_2}{[H]_1}\right)^{\alpha}$

$8 = 2^{\alpha} \cdot 2^{\beta}$ .....(1)

$2 = 2^{\alpha}$ .....(2)

$\alpha = 1, \beta = 2$

13. Under the same reaction conditions, initial concentration of 1.386 mol dm^{-3} of a substance becomes half in 40 seconds and 20 seconds through first order and zero order kinetics, respectively. Ratio $\left(\frac{k_1}{k_0}\right)$ of the rate constants for first order (k_1) and zero order (k_0) of the reactions is **[JEE 2008]**

(A) 0.5 mol^{-1} dm^3 (B) 1.0 mol dm^{-3}

(C) 1.5 mol dm^{-3} (D) 2.0 mol^{-1} dm^3

Ans. **(A)**

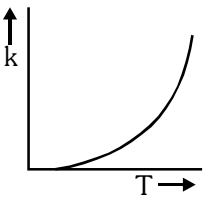
Sol. $K_1 = \frac{\ln 2}{40}$

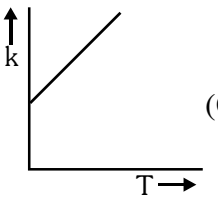
$K_0 = \frac{1.386}{2 \times 20}$

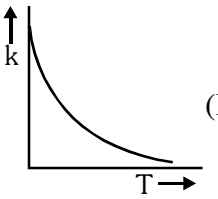
$\frac{K_1}{K_0} = 0.5$

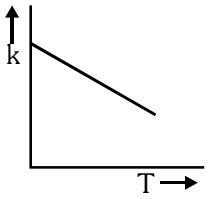
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14. Plots showing the variation of the rate constant (k) with temperature (T) are given below. The plot that follows Arrhenius equation is – **[JEE 2010]**

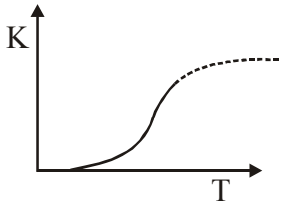
(A) 

(B) 

(C) 

(D) 

Ans. **(A)**

Sol. 

15. The concentration of R in the reaction R → P was measured as a function of time and the following data is obtained :

[R] (molar)	1.0	0.75	0.40	0.10
t(min.)	0.0	0.05	0.12	0.18

The order of the reaction is. **[JEE 2010]**

Ans. **Zero**

Sol. By hit and trial method het reaction is zero order.

$$K_1 = \frac{[A]_0 - [A]_1}{t_1}$$

$$K_1 = \frac{1 - 0.75}{0.05} = 5$$

$$K_2 = \frac{[A]_0 - [A]t_2}{t_2}$$

$$K_2 = \frac{1 - 0.4}{0.12} = 5$$

As $K_1 \approx K_1$ so reaction will be zero order reaction.

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